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Prevalence of Stress, Anxiety, Depression, and Sleep Quality Among Young Adults in Türkiye: A Cross-Sectional Study

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ABSTRACT

Aim: The current study was conducted to measure the prevalence of stress, anxiety, depression, and sleep quality and identify predisposing factors of psychological distress among young adults during the pandemic.

Methods: A cross-sectional design was adopted. The target population consisted of students studying at an associate degree health services school at a university in Türkiye. Overall, 704 students were included in the sample. Multiple logistic regression was implemented to predict risk factors.

Results: The prevalence of moderate to extremely severe stress, anxiety, and depression was found to be 39%, 48.4% and 47.6%, respectively. There was a strong positive association between stress and anxiety ($r = 0.869$, $p < 0.001$), stress and depression ($r = 0.912$, $p < 0.001$), and between anxiety and depression ($r = 0.857$, $p < 0.001$). A moderate positive relationship was found between sleep quality and stress ($r = 0.484$, $p < 0.001$), sleep quality and anxiety ($r = 0.484$, $p < 0.001$), and sleep quality and depression ($r = 0.481$, $p < 0.001$). Young adults with poor sleep quality, those who find safety measures for the pandemic outbreak insufficient, those who use alcohol, and those who feel despair were more likely to have stress, anxiety, and depression risk. Age and gender did not affect their stress, anxiety, or depression risk.

Conclusions: Nearly half of the young adults experienced varying degrees of stress, anxiety, and depression, and most had poor sleep quality. The insufficiency of outbreak measures, poor sleep quality, using alcohol, and feelings of despair were mutual predictive factors of stress, anxiety, and depression.

1 | Introduction

Depressive and anxiety disorders are among the most common psychiatric illnesses affecting young people. Anxiety disorders typically begin in childhood, whereas the onset of depression frequently occurs later, during adolescence or early adulthood [1]. Mental disorders were the leading causes of the 'global burden of disease', and depressive and anxiety disorders were the leading contributors to this burden before 2020 [2]. In the pre-pandemic period, the prevalence of mental problems in the young population was similar across the world [3–5]. Findings from New Zealand,

the Dutch, and the United States demonstrated that the 12-month prevalence was between 19.4% and 22.3% for anxiety disorders and between 8.3% and 12.4% for depressive disorders among young people [3]. A population-based survey in China revealed that mild to very severe prevalence of depression, anxiety, and stress were 27.3%, 33.4% and 12%, respectively, among college students [4]. Of the Spanish students, 23.6%, 34.5% and 18.4% had symptoms of anxiety, stress, and depression (mild to extremely severe), respectively [5]. Nevertheless, the percentages of mild to extremely severe levels of depression, anxiety, and stress were reported to be 39%, 47.8%, and 41.2% in Turkish university students [6]. Moreover, the

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emergence of the COVID-19 pandemic has created an environment where many determinants of impaired mental health are exacerbated [2]. The World Health Organization (2022) reported that the pandemic has affected the mental health of young people and that the global prevalence of anxiety and depression increased by a massive 25% in the first year of the COVID-19 pandemic [7].

Although young people are reportedly at lower risk for morbidity and mortality from coronavirus disease than individuals in other age groups, the effects of the COVID-19 pandemic on the mental health of youth have been significant [8]. A meta-analysis revealed that one in four adolescents globally were experiencing depression symptoms, while one in five were experiencing anxiety symptoms in the first year of the COVID-19 pandemic [9]. A highly severe prevalence of stress (40.7%), anxiety (44.7%), and depression (34.6%) was observed during the second year of the pandemic among university students [10]. In the third year of the pandemic, a decline in the extremely severe levels of stress, anxiety, and depression was reported [11]. A recent systematic review and meta-analysis consisted of college students from different countries such as China, Bangladesh, Spain, Russia, Korea, United States, and Japan revealed that the prevalence of anxiety, depression, and stress during the COVID-19 pandemic were 29%, 37% and 23%, respectively [12]. In Türkiye, the prevalence of high stress, high generalised anxiety, and high depression symptoms in university students were 84.2%, 36.2% and 55%, respectively [13]. Therefore, promoting mental health among this population is thought to be of importance to support their overall health and well-being.

Traumatic events, like the pandemic, may cause a physiologic stress response that can influence sleep quality and lead to sleep disorders [14]. Before the COVID-19 pandemic, studies conducted in different countries, including Brazil, China, Saudi Arabia, and Türkiye, the prevalence of poor sleep quality was reported between 31% and 65% among college students [15–18]. Moreover, studies have shown that insufficient sleep and poor sleep quality are closely related to mental health problems among college students [19, 20]. Poor sleep quality and sleep deprivation affect the quality of life and lead to mental problems such as stress, depression, and anxiety symptoms [20]. Furthermore, young people with poor sleep quality experience approximately five times more stress than those with good sleep quality before the pandemic [21]. More recently, pandemic measures contributed to sleep problems, sleep latency, and shorter sleep duration, which led to a substantial worsening of sleep quality among undergraduate students [22].

Changes and restrictions in daily life, such as prolonged home isolation, online learning, and social distancing due to the COVID-19 pandemic, have increased stress, anxiety, and depression levels, as well as affected sleep patterns and quality [23, 24]. A study by Ulrich et al. reported that 42% of college students had poor sleep quality, with approximately 60% reporting at least one sleep problem [25]. Reduced sleep quantity and quality during the pandemic were related to poor mental health outcomes [26]. In addition, poor sleep quality has a significant mediating effect in the relationship between stress and depression [24]. Several studies conducted during the first lockdown period reported increased sleep difficulties in

association with high levels of depression, anxiety, and stress among university students [27, 28].

The COVID-19 pandemic seemingly hurt both the sleep quality and the mental health of young adults [11, 25]. Moreover, alterations in daily life habits and educational processes, loneliness, fear of being infected, suffering and death for oneself and loved ones, grief after bereavement, and financial worries may have led to serious psychological impacts on the young population [7, 29]. Therefore, it is estimated that early diagnosis of mental health problems and a better understanding of high risk would contribute to the planning of timely interventions for young people and thus help decrease mental issues.

The current study was performed to measure the prevalence of stress, anxiety, depression, and sleep quality and identify predisposing factors of psychological distress among young adults during the pandemic. The research question was—‘What is the prevalence of stress, anxiety, depression, and sleep quality during the pandemic among young adults?’ The research hypothesis was that individual characteristics might affect stress, anxiety, and depression levels in young adults.

2 | Methods

2.1 | Study Design and Participants

A cross-sectional design was adopted. The target population was comprised of freshman and senior students studying at an associate degree health services school of a university in Türkiye ($N = 1050$). The estimated sample size was calculated using power analysis. Accordingly, the expected frequency was assumed to be 50%, and the minimum sample size was calculated as 297 based on a 5% deviation and 95% confidence interval. Eligible participants included all students who enrolled in the school-aged 18 years and over. Because of the increase in COVID-19 cases in Türkiye, distance education was started instead of face-to-face learning at the end of March 2020. The convenience sampling method was applied for data collection with an online questionnaire. Therefore, all students were invited to participate in the study by email and those who met the inclusion criteria were recruited. Overall, 810 students were included in the sample. The study was completed by 704 students.

The inclusion criteria were: (1) being 18 years or over, (2) the ability to complete the online questionnaire, and (3) agreeing to participate in the study. The exclusion criteria were: (1) being an international student, (2) having an audio-visual disability, and (3) having any neuropsychiatric problems. International students were excluded because they could have difficulties understanding the Turkish instruments. Individuals with audio-visual or neuropsychiatric problems may experience communication problems.

2.1.1 | Setting

The study was conducted at an associate degree health services school of a state university. The health services school programmes include medical documentation and secretary, medical

promotion and marketing, opticianry, elderly care, home care, child development, first and emergency aid, physiotherapy, disabled care, and rehabilitation. Students take intensive theoretical and practical courses during their 2 years of education.

The emergence of the COVID-19 pandemic has caused changes in the school setting and learning methods at the country level. The duration of the programmes carried out in the school, where the study was conducted, was 2 years, the interruption of face-to-face education due to the pandemic, the inadequacy of equipment such as computers and the internet during online education, and the lack of necessary infrastructure for departments with skill-based requirements, all negatively affected the quality of teaching and caused extra stress and anxiety in students. Moreover, academic performance, graduation, and employment pressures and changes in post-graduation plans may contribute to the exacerbation of mental health problems among students.

2.2 | Instruments

The data were collected by a participant information form based on the literature [28, 30], the Pittsburgh Sleep Quality Index (PSQI), and the Depression Anxiety Stress Scale (DASS). The participant information form contained nine questions such as age, gender, income, existence of chronic disease, medication use, smoking, drinking alcohol, emotions experienced during COVID-19, and their comfort level in the sufficiency of epidemic measures.

The DASS, developed by Lovibond and Lovibond [31] in 1995, consists of depression, anxiety, and stress sub-scales with a total of 42 items. The scale has 4 - Likert ratings: strongly disagree = 0, slightly agree = 1, usually agree = 2, and strongly agree = 3. Total scores of the scale range between 0 and 42 for each sub-scale. High scores obtained from each of the sub-scales of depression, anxiety, and stress reveal that the person has the relevant trait. The Turkish validity and reliability study was done by Akın and Çetin [32] in 2007, and the Cronbach Alpha (α) coefficients were found to be 0.89 for the whole scale. The Cronbach alpha was found to be 0.97 for this study.

The PSQI developed by Buysse et al. [33] was used to assess sleep quality and disturbances in the last month. The Turkish version adaptation was performed by Ağargün et al. [34] in 1996, and the Cronbach's Alpha coefficient was 0.80. Eighteen items are included in the scoring for the evaluation of the PSQI, which consists of 24 questions in total. The scale has seven subgroups, including subjective quality, latency and duration of sleep, usual sleep efficacy, sleep disturbances, usage of sleeping medication, and daytime dysfunction. Each subgroup is evaluated on a score of 0–3. The total score from the subgroups ranges from 0 to 21. The total PSQI score between 0 and 4 indicates good sleep quality, and scores between 5 and 21 show poor sleep quality.

2.3 | Procedure and Measurements

The study was conducted between April and June 2020. Because of the increase in COVID-19 cases in Türkiye, online

learning continued during the data collection period. Instruments were prepared as an online questionnaire using Google Forms. The student emails were obtained from the school management. Since the face-to-face learning has been interrupted, an online survey link was shared with the participants through email. The online data collection tools were organised as the questions could be reached following the approval of voluntary participation on the first page. After the voluntary students consented to participating, they completed the questionnaire and submitted their answers. The 'student ID' was added to the survey to prevent repeated participation. Researchers sent online reminders via emails twice a week to increase student participation. The complementation of the tools lasted about 15–20 min.

2.4 | Data Analysis

Data were assessed using the IBM SPSS (Statistical Package of Social Sciences) for Windows version 25.0 programme. The independent variables were the students' descriptive characteristics, while the dependent variables were depression, anxiety, and stress. Number, percentage, mean, and standard deviation (SD) were used to present descriptive characteristics. Simple and multiple logistic regression was performed with dichotomous dependent variables of stress, anxiety, and depression. Missing data, extreme value, autocorrelation, and multicollinearity were examined before the data analysis, and no violation was detected. The potential confounding factors were managed in the analysis stage using multivariate logistic regression models. The multivariable logistic regression analysis included significant independent variables ($p < 0.05$). Pearson's correlation coefficient was used for correlation. The significance level was accepted as $p < 0.05$.

2.5 | Ethical Consideration

Ethical approval was obtained from the Bartın University Social and Human Sciences Ethics Committee (2020-SBB-0081). Before implementing the instruments, participants were informed about the study. Informed consent was obtained by adding the 'I agree/disagree to participate in the study' section to the electronic questionnaire. Students were informed on the first page of the survey that they had the right to leave at any time and that their participation was voluntary. They were explained that the results of the study would not affect their course grades. The 'student ID' was added to the survey to protect their anonymity. Within the scope of the study, the confidentiality of the answers of the students was ensured. Student IDs were not shared with people other than the researchers.

3 | Results

Among the recruited students ($n = 810$), 62 refused the invitation, and 44 incorrectly completed the instruments. The final analysis was performed with 704 young adults. The response rate was 86.9%.

Of the young adults, 77% were in the 20–21 age group, 50.7% were first year students, 67.9% were female, 60.9% had income equal to the expenses, 91.2% had no chronic diseases, 89.8% did not use medicines, 32.7% smoked, 15.3% used alcohol, 62.9% experienced uncertainty, 39.6% experienced fear during the COVID-19 pandemic, 55.1% found epidemic measures sufficient, and 67.5% had poor sleep quality (Table 1). According to the total scores of the young adults from the DASS scale, the prevalence of moderate to extremely severe stress, anxiety, and depression was found to be 39%, 48.4% and 47.6%, respectively (Table 2).

TABLE 1 | Individual characteristics of young people ($N = 704$).

Individual characteristics	<i>n</i>	%
Age		
18–19 years	162	23.0
20–21 years	542	77.0
Academic level		
Freshman (first year)	357	50.7
Senior (second year)	347	43.3
Gender		
Female	478	67.9
Male	226	32.1
Income level		
Income less than expenses	220	31.3
Income equal to/more than expenses	484	68.7
Having chronic disease		
Yes	62	8.8
No	642	91.2
Regular medication use		
Yes	72	10.2
No	632	89.8
Smoking		
Yes	230	32.7
No	474	67.3
Using alcohol		
Yes	108	15.3
No	596	84.7
Emotions experienced during the COVID-19 ^a		
Uncertainty	443	62.9
Worry	430	61.1
Fear	279	39.6
Despair	184	26.1
Sufficiency of epidemic measures		
Yes	316	44.9
No	388	55.1
Sleep quality		
Normal	229	32.5
Poor	475	67.5

^aMultiple options were marked.

TABLE 2 | Depression, anxiety, and stress levels of young people ($N = 704$).

Level	Stress <i>n</i> (%)	Anxiety <i>n</i> (%)	Depression <i>n</i> (%)
Normal	357 (50.7)	311 (44.2)	287 (40.8)
Mild	72 (10.2)	53 (7.5)	82 (11.6)
Moderate	108 (15.3)	130 (18.5)	142 (20.2)
Severe	102 (14.5)	80 (11.4)	73 (10.4)
Extremely severe	65 (9.2)	130 (18.5)	120 (17.0)

Predisposing factors for stress were assessed using multiple logistic regression analysis. According to this analysis, low-income level (OR = 1.69, 95% CI: 1.18–2.44), having chronic disease (OR = 2.98, 95% CI: 1.54–5.78), smoking (OR = 1.61, 95% CI: 1.10–2.36), drinking alcohol (OR = 1.69, 95% CI: 1.01–2.82), feeling despair (OR = 2.42, 95% CI: 1.61–3.65), finding safety measures for the outbreak insufficient (OR = 1.52, 95% CI: 1.09–2.13), and poor sleep quality (OR = 3.77, 95% CI: 2.61–5.46) were predisposing factors for stress. There was no relationship between stress level and potential confounding factors, including uncertainty, worry, and fear ($p > 0.05$) (Table 3).

Risk factors for anxiety were assessed via multiple logistic regression analysis. Having chronic disease (OR = 2.37, 95% CI: 1.23–4.57), drinking alcohol (OR = 1.83, 95% CI: 1.09–3.09), feeling despair (OR = 2.04, 95% CI: 1.37–3.06), finding safety measures for the outbreak insufficient (OR = 1.61, 95% CI: 1.16–2.25), and poor sleep quality (OR = 4.41, 95% CI: 2.61–5.46) were risk factors for anxiety. The potential confounding factors, including uncertainty, worry, and fear, were not associated with anxiety level ($p > 0.05$) (Table 4).

Predisposing factors for depression were assessed by multiple logistic regression analysis. Low-income level (OR = 2.30, 95% CI: 1.57–3.35), using medication (OR = 3.23, 95% CI: 1.65–6.35), drinking alcohol (OR = 2.68, 95% CI: 1.60–4.49), feeling despair (OR = 2.60, 95% CI: 1.71–3.93), finding safety measures for the outbreak insufficient (OR = 1.50, 95% CI: 1.07–2.10), and poor sleep quality (OR = 3.21, 95% CI: 2.26–4.56) were predisposing factors for depression. The potential confounding factors, including uncertainty, worry, and fear, did not affect the depression level ($p > 0.05$) (Table 5).

A strong positive association was found between stress and anxiety ($r = 0.869$, $p < 0.001$), between stress and depression ($r = 0.912$, $p < 0.001$), and between anxiety and depression ($r = 0.857$, $p < 0.001$). A moderate positive relationship was found between sleep quality and stress ($r = 0.484$, $p < 0.001$), sleep quality and anxiety ($r = 0.484$, $p < 0.001$), and sleep quality and depression ($r = 0.481$, $p < 0.001$). In addition, age, academic level, and gender did not affect stress, anxiety, and depression in young adults.

4 | Discussion

The coronavirus outbreak has become a vital public health problem. Emergencies such as the COVID-19 pandemic can lead to emotional alterations among individuals [9, 35]. The

TABLE 3 | Predisposing factors associated with stress ($N = 704$).

Variables	<i>n</i>	Univariable analysis			Multiple analysis		
		OR	95% CI	<i>p</i>	OR	95% CI	<i>p</i>
Academic level							
Freshman (first year)	357	1.02	0.76–1.38	0.876		—	
Senior (second year)	347	1					
Gender							
Male	226	1				—	
Female	478	1.28	0.93–1.75	0.130			
Income level							
Less than expenses	220	1.83	1.32–2.52	< 0.001	1.69	1.18–2.44	0.005
Equal to/more than expenses	484	1			1		
Having chronic disease							
Yes	62	2.74	1.54–4.83	0.001	2.98	1.54–5.78	0.001
No	642	1			1		
Using medication							
Yes	72	2.76	1.62–4.69	< 0.001	1.42	0.67–2.98	0.357
No	632	1			1		
Smoking							
Yes	230	1.80	1.31–2.48	< 0.001	1.61	1.10–2.36	0.015
No	474	1			1		
Using alcohol							
Yes	108	1.93	1.26–2.94	0.002	1.69	1.01–2.82	0.045
No	596	1			1	1.01–2.80	
Worry							
Yes	430	1.16	0.85–1.56	0.349		—	
No	274	1					
Fear							
Yes	279	1.84	1.35–2.49	< 0.001	1.34	0.93–1.92	0.115
No	425	1			1		
Despair							
Yes	184	3.24	2.66–4.66	< 0.001	2.42	1.61–3.65	< 0.001
No	520	1			1		
Uncertainty							
Yes	443	1.39	1.02–1.89	0.033	1.40	0.99–1.98	0.060
No	261	1					
Sufficiency of measures							
Yes	316	1			1		
No	388	1.81	1.34–2.45	< 0.001	1.52	1.09–2.13	0.014
Sleep quality							
Poor	475	4.58	3.22–6.50	< 0.001	3.77	2.61–5.46	< 0.001
Good	229	1			1		

Abbreviations: CL, Confidence Interval; OR, Odds Ratio.

present study revealed about half of young adults experienced moderate to extremely severe levels of stress, anxiety, and depression. A recent systematic review and meta-analysis, including 104 studies, found that the pooled prevalence of depression was 32%, anxiety was 28%, and stress was 31%

among the student population during the pandemic [23]. A global survey conducted in different regions such as Australia, India, the United Kingdom, South Africa, Spain, and the United States reported that over 70% of the respondents had greater than moderate levels of stress, 59% had clinically

TABLE 4 | Predisposing factors associated with anxiety ($N = 704$).

Variables Academic level	<i>n</i>	Univariable analysis			Multiple analysis		
		OR	95% CI	<i>p</i>	OR	95% CI	<i>p</i>
Freshman (first year)	357	0.95	0.70–1.28				
Senior (second year)	347	1					
Gender							
Male	226	1					
Female	478	1.21	0.88–1.66	0.245			
Income level							
Less than expenses	220	1.83	1.32–2.52	< 0.001	1.38	0.96–1.98	0.081
Equal to/more than expenses	484	1			1		
Having chronic disease							
Yes	62	2.24	1.25–3.99	0.006	2.37	1.23–4.57	0.010
No	642	1			1		
Using medication							
Yes	72	2.40	1.39–4.14	0.002	0.30	0.60–2.78	0.499
No	632	1			1		
Smoking							
Yes	230	1.73	1.25–2.40	0.001	1.41	0.96–2.06	0.080
No	474	1			1		
Using alcohol							
Yes	108	2.09	1.34–3.25	0.001	1.83	1.09–3.09	0.023
No	596	1			1		
Worry							
Yes	430	1.34	0.98–1.81	0.063			
No	274	1					
Fear							
Yes	279	1.72	1.26–2.34	0.001	1.34	0.92–1.92	0.115
No	425	1			1		
Despair							
Yes	184	2.52	1.75–3.62	< 0.001	2.04	1.37–3.06	< 0.001
No	520	1			1		
Uncertainty							
Yes	443	1.21	0.89–1.64	0.227			
No	261	1					
Sufficiency of measures							
Yes	316	1			1		
No	388	1.86	1.37–2.51	< 0.001	1.61	1.16–2.25	0.005
Sleep quality							
Poor	475	5.13	3.64–7.23	< 0.001	4.41	3.09–6.29	< 0.001
Good	229	1			1		

Abbreviations: CL, Confidence Interval; OR, Odds Ratio.

probable anxiety, and 39% had moderate depressive symptoms [24]. In Türkiye, high perceived stress, moderate generalised anxiety disorder, and mild depression symptoms were found during the second wave of the pandemic among university students [13].

Studies conducted before the COVID-19 pandemic reported an even lower prevalence of psychological distress, including depression, anxiety, and stress in college students [4, 5]. It appears that there is a higher rate of mental problems in the young population during the COVID-19 pandemic compared to the pre-pandemic.

TABLE 5 | Predisposing factors associated with depression ($N = 704$).

Variables	<i>n</i>	Univariable analysis			Multiple analysis		
		OR	95% CI	<i>p</i>	OR	95% CI	<i>p</i>
Academic level							
Freshman (first year)	357	1.11	0.82–1.50			—	
Senior (second year)	347	1					
Gender							
Male	226	1				—	
Female	478	1.08	0.78–1.49	0.638			
Income level							
Less than expenses	220	2.43	1.71–3.44	< 0.001	2.30	1.57–3.35	< 0.001
Equal to/more than expenses	484	1			1		
Having chronic disease							
Yes	62	2.10	1.16–3.79	0.014	1.02	0.43–2.44	0.966
No	642	1			1		
Using medication							
Yes	72	3.47	1.87–6.46	< 0.001	3.23	1.65–6.35	0.001
No	632	1			1		
Smoking							
Yes	230	1.58	1.14–2.16	0.006	1.18	0.81–1.74	0.387
No	474	1			1		
Using alcohol							
Yes	108	2.446	1.54–3.93	< 0.001	2.68	1.60–4.49	< 0.001
No	596	1			1		
Worry							
Yes	430	0.91	0.67–1.24	0.560		—	
No	274	1					
Fear							
Yes	279	1.48	1.08–2.02	0.014	1.00	0.69–1.44	0.991
No	425	1			1		
Despair							
Yes	184	3.01	2.05–4.27	< 0.001	2.60	1.71–3.93	< 0.001
No	520	1			1		
Uncertainty							
Yes	443	1.21	0.89–1.65	0.228		—	
No	261	1					
Sufficiency of measures							
Yes	316	1			1		
No	388	1.82	1.34–2.47	< 0.001	1.50	1.07–2.10	0.018
Sleep quality							
Poor	475	3.82	2.74–5.32	< 0.001	3.21	2.26–4.56	< 0.001
Good	229	1			1		

Abbreviations: CL, Confidence Interval; OR, Odds Ratio.

Profound changes and restrictions have occurred in the daily lives of young adults due to the COVID-19 pandemic. Reduced learning activities in online education, prolonged lack of social activities, postponement of relevant professional exams, delayed academic

progress, and pressure to graduate may cause psychological distress [23]. It is thought that these unexpected situations that emerged with the pandemic caused a significant increase in stress, anxiety, and depression levels in young adults.

The pandemic affects not only individuals' emotional health but also their sleep quality [25]. The current study determined that most (67.5%) young adults have poor sleep quality. A global survey conducted in over 60 countries revealed that poor sleep quality was common among the young population, reported by 73% of the respondents [24]. A study conducted in Türkiye found that approximately 90% of university students were affected by the COVID-19 pandemic in different ways, and this caused a deterioration in sleep quality [36]. The prevalence of poor sleep quality in the present study was higher than in pre-pandemic studies [18, 28]. It is thought that the adverse influences on the psychological state of young adults during the pandemic, profound changes and restrictions in their life routines, and the unknown course and outcome of the pandemic may negatively affect their sleep quality.

Sleep quality is essential for the immune system to resist epidemic diseases such as COVID-19 and maintain health [37]. In this study, a moderate positive relationship was found between stress, anxiety, depression, and sleep quality. This finding shows that sleep quality decreases as young adults' stress, anxiety, and depression levels increase. In addition, those with poor sleep quality were found to have three to four times higher risk factors for stress, anxiety, and depression. Çıtak and Pekdemir [37] noted that adults who had sleep problems during the COVID-19 pandemic had a higher level of anxiety. Bulut et al. [36] reported that sleep quality was positively related to anxiety, depression, and insomnia severity. Zhou et al. [19] also reported that depression or anxiety was a risk factor for insomnia symptoms and that insomnia symptoms increased the symptoms of anxiety and depression. It is thought that both the psychological state and the sleep quality of the individual should be improved to protect and maintain their health. Following the stress caused by the pandemic, any sleep disturbance can increase exposure to infection, or the disease state may also put recovery at risk [38].

Smoking is associated with the higher frequency and increased severity of the clinical course of COVID-19 disease [39, 40]. In this study, one out of every three young adults smoked, and smoking was one of the predictors of stress and anxiety. Similarly, a study conducted in the United Kingdom reported that smokers who were anxious and worried about their mental health smoked more during the pandemic [39]. A 2-year longitudinal study conducted in Norway revealed a higher prevalence of smokers among individuals experiencing severe psychological distress compared to those without distress [41]. An Australian study found that those who reported an adverse change in smoking were more likely to experience stress, anxiety, and depression [38]. The high risk of stress and anxiety in young adults who smoke can be explained by the fact that young adults think that smokers are more at risk for COVID-19 disease, and they do not know what the clinical course of the process will be if diagnosed with the disease [39, 41]. Therefore, it is thought that the awareness of young adults about smoking and its adverse effects should be increased, and smoking cessation efforts should be planned.

The prevention measures taken during the pandemic can protect mental health in the early stages [11]. This study determined that those who believed that epidemic measures

were insufficient had a higher risk of stress, anxiety, and depression. Wang et al. [42] determined that wearing a mask and ensuring hand hygiene reduced the level of anxiety and depression. Zhou et al. [19] found that COVID-19 awareness (knowledge, prevention, and measures) protects against depressive and anxiety symptoms. It is thought that measures such as wearing masks, hand hygiene, and complying with social distancing to prevent the spread of COVID-19 positively affect the psychological health of young adults.

4.1 | Strengths and Limitations

The strength of the study is that the regression analysis results, which provide information about predictive factors, could help design interventions to improve the mental health of young adults during or after public health emergencies. The second strength of the study is that the data were collected in the first wave of the pandemic, and the early effects of the pandemic on mental health were revealed. Finally, the present study was conducted with a large sample and had a reasonable response rate.

The study is limited to the data obtained from students with internet access who participate in the electronic survey. All instruments were self-reported questionnaires, resulting in the possibility of response bias. Due to the closed-ended questions enclosed in the instruments, situations that could affect mental status could not be addressed in detail. The exact prevalence of stress, anxiety, and depression cannot be estimated. The exact causal relationships could be limited due to the cross-sectional design. Due to the convenience sampling method, which was not based on a random selection, the possibility of sampling bias should be considered. Since the data were collected during the home quarantine period, familial and environmental factors may have influenced the emotional status and sleep patterns of the participants. Moreover, potential confounding factors, including the risk and fear of being infected with the virus, uncertainty about the daily routine and the treatment methods, and being socially isolated, could have affected the mental status and sleep patterns among the participants. Although the depression, anxiety, and stress scales used in this study are valid and reliable, they may not be as sufficient as clinical assessment. Since the participants were studying at different health-related programmes, perceptions and experiences of stress, anxiety, and depression may have differed.

4.2 | Implications for Practice

Health professionals, especially nurses and mental health counsellors, could implement interventions to decrease the adverse consequences of mental health issues and improve sleep quality among young adults. Future studies could focus on effective methods, such as relaxation exercises, mindfulness that strengthen mental health and improve coping skills in situations such as a pandemic, and environments or practices that increase sleep quality, and qualitative studies to ensure better an understanding of the relationship between academic life, mental health, and sleep quality. By using education and counselling roles, nurses could plan activities to improve the mental health of the community, especially young adults.

For example, nurses could organise community-based education programmes to enhance coping skills and increase the psychological resilience of individuals during health emergencies. Nurses could support individuals by providing resources to improve their health literacy skills, help them distinguish high and low-quality resources on the internet, and assist them in evaluating reliable health resources. Since public health emergencies, such as the COVID-19 pandemic, may affect individuals all over the world, experimental studies could be designed to cope with stress, prevent anxiety and depression, and improve sleep quality in young adults. Moreover, studies could be designed to compare stress, anxiety, depression levels, and sleep quality of students studying in health-related departments and other academic disciplines, including science versus humanities, and identify the risk factors of these problems among university students.

5 | Conclusion

The study concluded that approximately half of the young adults experienced varying degrees of stress, anxiety and depression, and most had poor sleep quality. The present study found a moderate positive relationship between the depression, anxiety and stress levels of young adults and their sleep quality during the pandemic. Moreover, poor sleep quality, feeling despair, insufficiency of epidemic measures, and using alcohol were found to be mutual predisposing factors of stress, anxiety, and depression. Based on these results, it is essential to address young adults' sleep patterns and mental health concerns and to combat the cognitive aspects of the pandemic. It is recommended to implement strategies for improving mental health and preventing the long-term effects of the pandemic on young adults.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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